10 Analytical Business Questions — Courier & Logistics DW

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Methodology: Ralph Kimball Dimensional Modeling Lifecycle

1. Executive Overview

In the **Kimball Dimensional Modeling Lifecycle**, analytical business questions dictate the boundaries, grains, and dimensional attributes of the enterprise data warehouse. Rather than executing a raw data dump from operational transactional databases, a Kimball warehouse is designed top-down from the analytical inquiries of business decision-makers, and bottom-up from the atomic operational events.

This document formalizes **10 core analytical business questions** across five strategic functional domains:

1. **Delivery Performance Analytics** (SLA adherence and operational execution)
2. **Customer Analytics** (Account value, churn prevention, and segment failure rates)
3. **Rider & Workforce Analytics** (Labor productivity, route coverage, and fleet balance)
4. **Hub & Regional Flow Analytics** (Sorting throughput, backlog dwell, and bottleneck diagnosis)
5. **Revenue & Financial Analytics** (Route contribution margins and cost-to-serve economics)

All target benchmarks and baseline figures cited below represent **assumed baseline figures for this case study**.

2. Business Questions Summary Matrix

ID	Thematic Group	Business Question Summary	Primary KPI	Source Fact Table(s)	Key Dimensions
BQ-01	Delivery Performance	What is our monthly on-time delivery rate across service tiers and regional hubs?	On-Time Delivery Rate (OTDR %)	`Fact_Delivery_Attempt`	Date, Service Type, Hub

BQ-02	Delivery Performance	What are the primary root-cause exception codes driving first-attempt delivery failures?	First-Attempt Delivery Rate (FADR %)	`Fact_Delivery_Attempt`	Delivery Status, Route, Customer
BQ-03	Customer Analytics	Which corporate B2B customer accounts generate high parcel volumes but suffer repeated SLA breaches?	Corporate SLA Adherence %	`Fact_Delivery_Attempt`	Customer, Date, Service Type
BQ-04	Customer Analytics	What is the delivery failure rate across corporate vs. individual customer segments by destination postal zone?	Segment Delivery Failure Rate (%)	`Fact_Delivery_Attempt`	Customer, Route, Delivery Status
BQ-05	Rider & Workforce	What is the average delivery productivity (Deliveries per Rider-Hour) across shifts and rider experience bands?	Deliveries per Rider-Hour (DPRH)	`Fact_Delivery_Attempt`, `Fact_Rider_Route_Assignment`	Rider, Shift, Date, Route
BQ-06	Rider & Workforce	Which delivery routes experience recurring unassigned route days and schedule coverage deficits?	Route Schedule Coverage Ratio (%)	`Fact_Rider_Route_Assignment`	Route, Hub, Date, Shift
BQ-07	Hub & Regional Flow	What is the daily end-of-day parcel backlog ratio and clearance rate across regional sorting hubs?	Hub Backlog Ratio (HBR %)	`Fact_Daily_Hub_Backlog`	Hub, Date, Parcel Type

BQ-08	Hub & Regional Flow	Which package weight categories and parcel types endure the longest average dwell latency at sorting centers?	Average Parcel Dwell Time (Hours)	`Fact_Daily_Hub_Backlog`	Parcel Type, Hub, Date
BQ-09	Revenue & Financial	Which delivery routes generate negative net contribution margins after factoring in fuel and rider labor?	Route Net Contribution Margin (RNCM $)	`Fact_Delivery_Attempt`	Route, Hub, Date, Service Type
BQ-10	Revenue & Financial	What is the financial cost of re-delivery attempts compared to the billed delivery fee across service tiers?	Re-Attempt Cost Ratio (%)	`Fact_Delivery_Attempt`	Service Type, Date, Delivery Status

3. Detailed Specifications: 10 Analytical Business Questions

Group 1: Delivery Performance Analytics

BQ-01: SLA Adherence & Lateness across Service Tiers and Regional Hubs

Exact Business Question: *"What is our monthly On-Time Delivery Rate (OTDR) across Same-Day, Express, and Standard service tiers, and which originating hubs fail to meet the 95.0% SLA contractual threshold?"*

Business Motivation & Decision Impact: Express and Same-Day services command premium pricing but carry contractual rebate penalties when delayed. This inquiry identifies chronic operational delays at the hub or regional level, allowing the Chief Operating Officer (COO) to adjust sorting schedules or renegotiate unachievable delivery windows.

Primary KPI & Target: On-Time Delivery Rate (OTDR %). Benchmark Target: >= 95.0% (*assumed baseline target for this case study; current baseline 82.0%*).
Dimensions & Attributes: `Dim_Date` (Year, Quarter, Month), `Dim_Service_Type` (Service_Tier_Name), `Dim_Hub` (Hub_Name, City, Region_Name).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT d.Year_Month, h.Region_Name, h.Hub_Name, s.Service_Tier_Name, COUNT(CASE WHEN
stat.Attempt_Outcome = 'Delivered' AND f.Attempt_Timestamp <= f.SLA_Deadline_Timestamp THEN 1 END)
AS On_Time_Deliveries, COUNT(CASE WHEN stat.Attempt_Outcome = 'Delivered' THEN 1 END) AS
Total_Deliveries, ROUND(100.0 * COUNT(CASE WHEN stat.Attempt_Outcome = 'Delivered' AND
f.Attempt_Timestamp <= f.SLA_Deadline_Timestamp THEN 1 END) / NULLIF(COUNT(CASE WHEN
stat.Attempt_Outcome = 'Delivered' THEN 1 END), 0), 2) AS OTDR_Percentage FROM Fact_Delivery_Attempt
f JOIN Dim_Date d ON f.Attempt_Date_Key = d.Date_Key JOIN Dim_Hub h ON f.Origin_Hub_Key = h.Hub_Key
JOIN Dim_Service_Type s ON f.Service_Type_Key = s.Service_Type_Key GROUP BY d.Year_Month,
h.Region_Name, h.Hub_Name, s.Service_Tier_Name;
```

Expected Output: Grouped bar chart showing monthly OTDR trend lines by service tier with 95% SLA target line.

BQ-02: First-Attempt Delivery Failure Patterns & Root-Cause Exceptions

Exact Business Question: *"What proportion of delivery attempts fail on the initial run, and what are the predominant exception failure reason codes across urban vs. suburban delivery routes?"*

Business Motivation & Decision Impact: Every failed delivery attempt requires returning the parcel to a hub, staging it for re-sort, and dispatching a secondary run. Uncovering failure reasons (e.g., "Customer Unavailable", "Incorrect Address", "Gate Locked") allows dispatchers to introduce proactive SMS notifications or require upfront address validation.

Primary KPI & Target: First-Attempt Delivery Rate (FADR %). Benchmark Target: >= 85.0% (*assumed baseline target; current baseline 76.0%*).
Dimensions & Attributes: `Dim_Delivery_Status` (Attempt_Outcome, Failure_Reason_Code), `Dim_Route` (Route_Code, Route_Type), `Dim_Date` (Day_Of_Week_Name).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT r.Zone_Classification, stat.Failure_Reason_Code, COUNT(f.Delivery_Attempt_Key) AS
Failed_Attempt_Count, ROUND(100.0 * COUNT(f.Delivery_Attempt_Key) /
SUM(COUNT(f.Delivery_Attempt_Key)) OVER(PARTITION BY r.Zone_Classification), 2) AS
Pct_Of_Route_Failures FROM Fact_Delivery_Attempt f JOIN Dim_Delivery_Status stat ON
f.Delivery_Status_Key = stat.Delivery_Status_Key JOIN Dim_Route r ON f.Route_Key = r.Route_Key WHERE
f.Attempt_Sequence_Number = 1 AND stat.Attempt_Outcome = 'Delivery Failed' GROUP BY
r.Zone_Classification, stat.Failure_Reason_Code;
```

Expected Output: 100% stacked bar chart depicting breakdown of failure exception categories across route types.

Group 2: Customer Analytics

BQ-03: Corporate Account Delivery Volume, Churn Risk & SLA Compliance

Exact Business Question: *"Which corporate B2B accounts generate the highest parcel volume, what is their SLA compliance rate over the past two quarters, and which accounts are at high risk of contract termination due to SLA degradation?"*

Business Motivation & Decision Impact: Corporate B2B clients contribute reliable recurring volume through scheduled daily pickups. However, enterprise accounts maintain strict SLA clawback penalty clauses. The Commercial Director needs a proactive scorecard to identify accounts suffering from poor delivery performance before contract renewal discussions.

Primary KPI & Target: Account SLA Compliance (%). Benchmark Target: >= 98.0% for Tier-1 Corporate Accounts (*assumed baseline target*).
Dimensions & Attributes: `Dim_Customer` (Customer_ID, Company_Name, Customer_Segment, Contract_Tier, Current_Row_Indicator), `Dim_Date` (Year, Quarter), `Dim_Service_Type` (Service_Tier_Name).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT c.Customer_ID, c.Company_Name, c.Contract_Tier, COUNT(f.Delivery_Attempt_Key) AS
Total_Deliveries, ROUND(100.0 * COUNT(CASE WHEN stat.Attempt_Outcome = 'Delivered' AND
f.Attempt_Timestamp <= f.SLA_Deadline_Timestamp THEN 1 END) / NULLIF(COUNT(f.Delivery_Attempt_Key),
0), 2) AS Corporate_SLA_Pct FROM Fact_Delivery_Attempt f JOIN Dim_Customer c ON f.Customer_Key =
c.Customer_Key JOIN Dim_Date d ON f.Attempt_Date_Key = d.Date_Key WHERE c.Customer_Segment = 'B2B
Enterprise' AND c.Current_Row_Indicator = 'Current' GROUP BY c.Customer_ID, c.Company_Name,
c.Contract_Tier HAVING COUNT(f.Delivery_Attempt_Key) >= 200;
```

Expected Output: Scatter bubble plot with Volume on X-axis, SLA % on Y-axis, and Bubble Size as Billed Revenue.

BQ-04: Customer Segment Failure Rates & Address Inaccuracy Analysis

Exact Business Question: *"How do delivery failure rates compare between B2B commercial accounts and B2C individual e-commerce recipients across major postal zones?"*

Business Motivation & Decision Impact: B2B deliveries occur during predictable office hours to registered docks, whereas B2C residential deliveries suffer from absent recipients and non-standard addresses. Quantifying this disparity enables commercial leadership to enforce differentiated delivery fee structures and residential surcharge adjustments.

Primary KPI & Target: Segment Delivery Failure Rate (%).
Dimensions & Attributes: `Dim_Customer` (Customer_Segment, City, Postal_Code), `Dim_Route` (Route_Code, Zone_Classification), `Dim_Delivery_Status` (Attempt_Outcome).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT c.Customer_Segment, r.Zone_Classification, COUNT(f.Delivery_Attempt_Key) AS Total_Attempts,
COUNT(CASE WHEN stat.Attempt_Outcome = 'Delivery Failed' THEN 1 END) AS Failed_Attempts, ROUND(100.0
* COUNT(CASE WHEN stat.Attempt_Outcome = 'Delivery Failed' THEN 1 END) /
NULLIF(COUNT(f.Delivery_Attempt_Key), 0), 2) AS Failure_Rate_Pct FROM Fact_Delivery_Attempt f JOIN
Dim_Customer c ON f.Customer_Key = c.Customer_Key JOIN Dim_Route r ON f.Route_Key = r.Route_Key JOIN
Dim_Delivery_Status stat ON f.Delivery_Status_Key = stat.Delivery_Status_Key GROUP BY
c.Customer_Segment, r.Zone_Classification;
```

Expected Output: Comparative paired bar chart contrasting failure percentages between B2B and B2C segments.

Group 3: Rider & Workforce Analytics

BQ-05: Rider Route Productivity (Deliveries per Rider-Hour) across Shifts

Exact Business Question: *"What is the average delivery productivity (Deliveries per Rider-Hour) achieved across morning vs. evening shifts, and how does rider tenure/vehicle type impact hourly throughput?"*

Business Motivation & Decision Impact: Workforce labor constitutes the single largest variable operating expense in courier logistics. Management requires an objective measure of rider productivity to benchmark route performance, set daily parcel dispatch quotas, and evaluate motorcycle vs. van delivery efficiency.

Primary KPI & Target: Deliveries per Rider-Hour (DPRH). Benchmark Target: >= 3.5 Deliveries / Hour (*assumed baseline target for this case study*).
Dimensions & Attributes: `Dim_Rider` (Rider_Name, Vehicle_Type, Employment_Type), `Dim_Shift` (Shift_Name, Standard_Shift_Hours), `Dim_Date` (Date_Value), `Dim_Route` (Route_Code).
Source Fact Table & Grain: Multi-Fact Drill-Across between `Fact_Rider_Route_Assignment` (pure factless keys) and `Fact_Delivery_Attempt` (transaction attempts) along conformed Date, Rider, and Route keys.

```
WITH Scheduled_Hours AS ( SELECT a.Rider_Key, a.Assignment_Date_Key AS Date_Key, a.Route_Key,
SUM(s.Standard_Shift_Hours) AS Total_Hours FROM Fact_Rider_Route_Assignment a JOIN Dim_Shift s ON
a.Shift_Key = s.Shift_Key GROUP BY a.Rider_Key, a.Assignment_Date_Key AS Date_Key, a.Route_Key ),
Completed_Deliveries AS ( SELECT f.Rider_Key, f.Attempt_Date_Key AS Date_Key, f.Route_Key,
COUNT(f.Delivery_Attempt_Key) AS Deliveries_Completed FROM Fact_Delivery_Attempt f WHERE
stat.Attempt_Outcome = 'Delivered' GROUP BY f.Rider_Key, f.Attempt_Date_Key, f.Route_Key ) SELECT
CONCAT(r.First_Name, ' ', r.Last_Name) AS Rider_Name, r.Vehicle_Type, SUM(cd.Deliveries_Completed)
AS Deliveries, SUM(sh.Total_Hours) AS Hours, ROUND(SUM(cd.Deliveries_Completed) /
NULLIF(SUM(sh.Total_Hours), 0), 2) AS Deliveries_Per_Rider_Hour FROM Scheduled_Hours sh JOIN
Completed_Deliveries cd ON sh.Rider_Key = cd.Rider_Key AND sh.Date_Key = cd.Date_Key AND
sh.Route_Key = cd.Route_Key JOIN Dim_Rider r ON sh.Rider_Key = r.Rider_Key GROUP BY
CONCAT(r.First_Name, ' ', r.Last_Name) AS Rider_Name, r.Vehicle_Type;
```

Expected Output: Box-and-whisker plot displaying distribution of Deliveries per Rider-Hour by Vehicle Type and Shift.

BQ-06: Fleet Route Coverage Discrepancies and Unassigned Route Days

Exact Business Question: *"Which delivery routes suffered from scheduled coverage deficits (unassigned route days) over the past month, and what is the correlation between unassigned routes and regional SLA breaches?"*

Business Motivation & Decision Impact: When a route has no scheduled rider assigned, parcels accumulate or must be absorbed by adjacent drivers, causing route delays. Identifying chronic unassigned route patterns enables dispatchers to recruit contractual cover drivers or reconfigure route boundaries.

Primary KPI & Target: Route Schedule Coverage Ratio (%).
Dimensions & Attributes: `Dim_Route` (Route_Code, Route_Name), `Dim_Hub` (Hub_Name, City), `Dim_Date` (Date_Value, Is_Working_Day_Flag).
Source Fact Table & Grain: `Fact_Rider_Route_Assignment` (Factless Fact). Grain: One row per scheduled assignment. Detected via outer join against calendar days.

```
SELECT h.Hub_Name, r.Route_Code, COUNT(d.Date_Key) AS Total_Working_Days, COUNT(a.Rider_Key) AS Assigned_Days, (COUNT(d.Date_Key) - COUNT(a.Rider_Key)) AS Unassigned_Route_Days, ROUND(100.0 * COUNT(a.Rider_Key) / NULLIF(COUNT(d.Date_Key), 0), 2) AS Route_Coverage_Pct FROM Dim_Route r CROSS JOIN Dim_Date d JOIN Dim_Hub h ON r.Assigned_Hub_Code = h.Hub_Code LEFT JOIN Fact_Rider_Route_Assignment a ON r.Route_Key = a.Route_Key AND d.Date_Key = a.Assignment_Date_Key WHERE d.Is_Working_Day_Flag = 'Yes' AND d.Year_Month = '2026-09' GROUP BY h.Hub_Name, r.Route_Code HAVING (COUNT(d.Date_Key) - COUNT(a.Rider_Key)) > 0;
```

Expected Output: Calendar heatmap showing unassigned route gap counts across days of the month.

Group 4: Hub & Regional Flow Analytics

BQ-07: Daily End-of-Day Parcel Backlog & Clearance Ratio Trends by Hub

Exact Business Question: *"What is the daily End-of-Day (EOD) parcel backlog count and clearance ratio for each regional distribution hub, and which hubs exceed operational holding thresholds?"*

Business Motivation & Decision Impact: Sorting hubs must process and dispatch all inbound packages within shift windows. Chronic parcel buildup indicates sorting bottlenecks or line-haul arrival delays. Historical snapshot analysis highlights which facilities require material handling automation or supplementary evening shifts.

Primary KPI & Target: Hub Backlog Ratio (HBR %). Benchmark Target: <= 8.0% (*assumed baseline target for this case study*).
Dimensions & Attributes: `Dim_Hub` (Hub_Name, Hub_Type, City, Region_Name), `Dim_Date` (Date_Value, Month_Name), `Dim_Parcel_Type` (Parcel_Category_Name).
Source Fact Table & Grain: `Fact_Daily_Hub_Backlog` (Periodic Snapshot Fact). Grain: One row per hub, date, parcel type, and backlog status at 23:59:59 daily.

```
SELECT d.Date_Value, h.Hub_Name, SUM(b.Backlog_Parcel_Count) AS Total_EOD_Backlog, SUM(b.Daily_Inflow_Count) AS Total_Inbound_Parcels, ROUND(100.0 * SUM(b.Backlog_Parcel_Count) / NULLIF(SUM(b.Daily_Inflow_Count), 0), 2) AS Hub_Backlog_Ratio_Pct FROM Fact_Daily_Hub_Backlog b JOIN Dim_Hub h ON b.Hub_Key = h.Hub_Key JOIN Dim_Date d ON b.Snapshot_Date_Key = d.Date_Key WHERE d.Date_Value >= DATEADD(day, -30, CURRENT_DATE()) GROUP BY d.Date_Value, h.Hub_Name;
```

Expected Output: Multi-series line chart tracking daily EOD Backlog Counts over time across regional hubs.

BQ-08: Regional Parcel Dwell-Time Latency by Parcel Type and Weight Class

Exact Business Question: *"What is the average dwell time of backlogged parcels sitting in distribution hubs, and which specific parcel weight and volume categories suffer the longest transit latency?"*

Business Motivation & Decision Impact: Standard flyer packages flow through sorting chutes rapidly, while oversized cartons require manual handling and endure extended dwell times. Identifying delays by package profile assists managers in planning physical staging areas and dedicated bulky transport.

Primary KPI & Target: Average Parcel Dwell Time (Hours). Benchmark Target: <= 6.0 Hours (*assumed baseline target for this case study*).
Dimensions & Attributes: `Dim_Parcel_Type` (Parcel_Category_Name, Weight_Class, Volumetric_Bucket), `Dim_Hub` (Hub_Name, Region_Name), `Dim_Date` (Year, Month_Name).
Source Fact Table & Grain: `Fact_Daily_Hub_Backlog` (Periodic Snapshot Fact). Grain: One row per hub, date, parcel type, and backlog status at 23:59:59 daily.

```
SELECT p.Parcel_Category_Name, p.Weight_Class, h.Region_Name, SUM(b.Backlog_Parcel_Count) AS Total_Staged_Parcels, ROUND(SUM(b.Total_Dwell_Hours) / NULLIF(SUM(b.Backlog_Parcel_Count), 0), 2) AS Average_Dwell_Hours FROM Fact_Daily_Hub_Backlog b JOIN Dim_Parcel_Type p ON b.Parcel_Type_Key = p.Parcel_Type_Key JOIN Dim_Hub h ON b.Hub_Key = h.Hub_Key JOIN Dim_Date d ON b.Snapshot_Date_Key = d.Date_Key GROUP BY p.Parcel_Category_Name, p.Weight_Class, h.Region_Name;
```

Expected Output: Heatmap matrix with Parcel Weight Classes on vertical axis, Regions on horizontal axis, and cell color as Dwell Hours.

Group 5: Revenue & Financial Analytics

BQ-09: Route Net Contribution Margin and Loss-Making Delivery Corridors

Exact Business Question: *"Which delivery routes yield negative or sub-target net contribution margins after deducting estimated fuel expenses and driver labor costs from billed delivery revenues?"*

Business Motivation & Decision Impact: Operating high-mileage suburban routes with low parcel drop densities often costs more in driver pay and van fuel than the revenue generated. Exposing loss-making routes allows the CFO to adjust regional boundaries, implement minimum stop fees, or renegotiate B2B contracts.

Primary KPI & Target: Route Net Contribution Margin (RNCM $). Benchmark Target: >= 22.0% Net Margin (*assumed baseline target*).
Dimensions & Attributes: `Dim_Route` (Route_Code, Route_Name, Zone_Classification), `Dim_Hub` (Hub_Name, City), `Dim_Date` (Year, Quarter), `Dim_Service_Type` (Service_Tier_Name).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT r.Route_Code, h.Hub_Name, r.Zone_Classification, COUNT(f.Delivery_Attempt_Key) AS Total_Deliveries, SUM(f.Delivery_Fee_Amount) AS Total_Revenue, SUM(f.Estimated_Labor_Cost + f.Estimated_Fuel_Cost) AS Operating_Costs, SUM(f.Net_Margin_Amount) AS Net_Margin_Dollars, ROUND(100.0 * SUM(f.Net_Margin_Amount) / NULLIF(SUM(f.Delivery_Fee_Amount), 0), 2) AS Net_Margin_Pct FROM Fact_Delivery_Attempt f JOIN Dim_Route r ON f.Route_Key = r.Route_Key JOIN Dim_Hub h ON f.Origin_Hub_Key = h.Hub_Key JOIN Dim_Date d ON f.Attempt_Date_Key = d.Date_Key WHERE stat.Attempt_Outcome = 'Delivered' AND d.Year = 2026 GROUP BY r.Route_Code, h.Hub_Name, r.Zone_Classification;
```

Expected Output: Diverging horizontal bar chart displaying positive margins in green and negative margins in red.

BQ-10: Cost of Re-delivery Attempts vs. Revenue Yield by Service Tier

Exact Business Question: *"What is the total operational cost incurred by repeated secondary and tertiary delivery attempts, and what percentage of billed delivery revenue is eroded by delivery failures across service tiers?"*

Business Motivation & Decision Impact: Delivery pricing assumes first-attempt completion. When shipments require 2 or 3 attempts, the marginal labor and fuel expenses erode profitability. Quantifying this margin leakage enables finance to institute re-attempt fee surcharges for chronic residential failure addresses.

Primary KPI & Target: Re-Attempt Cost Ratio (%).
Dimensions & Attributes: `Dim_Service_Type` (Service_Tier_Name), `Dim_Delivery_Status` (Attempt_Outcome, Failure_Reason_Code), `Dim_Date` (Quarter, Month_Name).
Source Fact Table & Grain: `Fact_Delivery_Attempt` (Transaction Fact). Grain: One row per physical delivery attempt.

```
SELECT s.Service_Tier_Name, COUNT(f.Delivery_Attempt_Key) AS Total_Attempt_Runs, COUNT(CASE WHEN
f.Attempt_Sequence_Number > 1 THEN 1 END) AS Secondary_Re_Attempts, SUM(CASE WHEN
f.Attempt_Sequence_Number = 1 THEN f.Delivery_Fee_Amount ELSE 0 END) AS Billed_Revenue, SUM(CASE
WHEN f.Attempt_Sequence_Number > 1 THEN (f.Estimated_Labor_Cost + f.Estimated_Fuel_Cost) ELSE 0 END)
AS Re_Attempt_Operating_Cost, ROUND(100.0 * SUM(CASE WHEN f.Attempt_Sequence_Number > 1 THEN
(f.Estimated_Labor_Cost + f.Estimated_Fuel_Cost) ELSE 0 END) / NULLIF(SUM(f.Delivery_Fee_Amount),
0), 2) AS Revenue_Erosion_Pct FROM Fact_Delivery_Attempt f JOIN Dim_Service_Type s ON
f.Service_Type_Key = s.Service_Type_Key JOIN Dim_Date d ON f.Attempt_Date_Key = d.Date_Key WHERE
d.Year = 2026 GROUP BY s.Service_Tier_Name;
```

Expected Output: Waterfall chart illustrating billed revenue eroded step-by-step by re-delivery attempt expenses.